21 · Content Projection and ViewChild

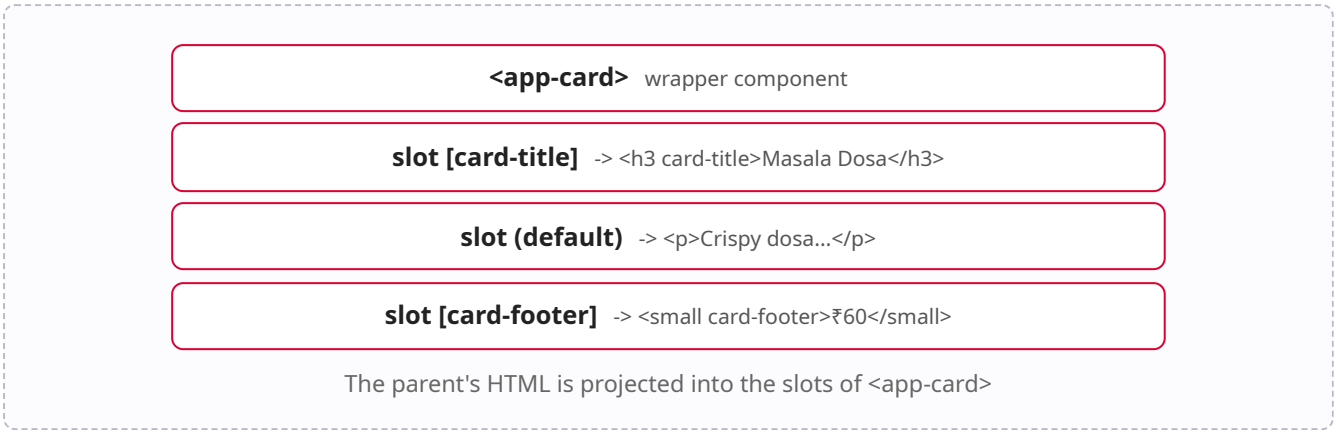
Intermediate Angular learning notes

a. Definition

Content projection lets a component accept HTML from its parent and place it in 'slots', using `<ng-content>`. It is how you build reusable wrappers such as cards, dialogs and tabs. Use the `select` attribute to create several named slots.

viewChild() gives a component a reference to an element or child component in its own template (marked with a template variable like `#box`). It is a signal, available once the view has rendered.

b. Diagram



c. Syntax

```
<!-- card.component template -->
<ng-content select="[card-title]" />
<ng-content />                        <!-- default slot -->

<!-- using it -->
<app-card><h3 card-title>Hi</h3> body text</app-card>

box = viewChild.required<ElementRef<HTMLInputElement>>('box');   // <input #box>
```

d. Example Program

Programs/21_Content_Projection_and_ViewChild/card-projection.component.ts

```
import { Component, ElementRef, signal, viewChild } from '@angular/core';

// A reusable wrapper with three slots
@Component({
  selector: 'app-card',
  template: `
    <div class="card">
      <header><ng-content select="[card-title]" /></header>
      <section><ng-content /></section>
      <footer><ng-content select="[card-footer]" /></footer>
    </div>
  `,
```

```
  styles: `.card { border: 1px solid #aaa; padding: 8px; border-radius: 6px; }`,
})
export class CardComponent {}

@Component({
  selector: 'app-root',
  imports: [CardComponent],
  template: `
    <app-card>
      <h3 card-title>Masala Dosa</h3>
      <p>Crispy dosa with potato filling.</p>
      <small card-footer>Price: ₹60</small>
    </app-card>

    <!-- #nameInput is a template reference variable used by viewChild -->
    <input #nameInput placeholder="Your name" />
    <button id="focus" (click)="focusInput()">Focus the box</button>
    <p>{{ status() }}</p>
  `,
})
export class AppComponent {
  nameInput = viewChild.required<ElementRef<HTMLInputElement>>('nameInput');
  status = signal('Not focused yet');

  focusInput() {
    const el = this.nameInput().nativeElement;
    el.focus();
    this.status.set(`Input focused: ${document.activeElement === el}`);
  }
}
```

e. Output

Rendered in the browser (text on screen, after the UI steps below):

1. Click **Focus the box**.

```
Masala Dosa
Crispy dosa with potato filling.
Price: ₹60
Focus the box
Input focused: true
```

f. Real-World Example (English + தமிழ்)

English

You buy an empty gift box: the shop gives the box, lid and ribbon, and you choose what to put inside. `<ng-content>` is the space for your items.

தமிழ்

கடையில் ஒரு காலிப் பரிசுப் பெட்டி வாங்குகிறீர்கள். பெட்டியின் வடிவம், மூடி, ரிப்பன் எல்லாம் கடைக்காரர் தருகிறார்; உள்ளே என்ன வைக்க வேண்டும் என்பது உங்கள் விருப்பம் - இனிப்போ, துணியோ, புத்தகமோ.

`<app-card>` அந்தப் பெட்டி; `<ng-content>` உள்ளே வைக்கும் இடங்கள் (மேலே தலைப்பு, நடுவில் பொருள், கீழே விலை). `viewChild` என்பது பெட்டிக்குள் இருக்கும் ஒரு

குறிப்பிட்ட பொருளைத் தொட்டு எடுப்பது போல.

g. Pros

- Flexible wrapper components
- Multiple slots with `select`
- Keeps components generic

h. Cons

- Projected content is created even if not shown
- Harder to type-check than inputs

i. Advanced / New Features

- Signal queries `viewChild()` / `contentChild()` (stable v19)
- `ngProjectAs`

j. Common Mistakes

- Reading `viewChild()` in the constructor or `ngOnInit`; it is only available after the view is created.
- Expecting `<ng-content>` to render conditionally; projected content is always created by the parent.
- Using `document.getElementById` instead of `viewChild`.
- Forgetting the `select` attribute so everything lands in the default slot.

k. Interview Questions

Q1. What is content projection?
Passing template content from a parent into a child component's `<ng-content>` slots.

Q2. What is `viewChild`?
A query that returns a child element/component from the component's own template.

Q3. `viewChild` vs `contentChild`?
`viewChild` queries the component's own template; `contentChild` queries projected content passed in by the parent.

Q4. What is a template reference variable?
A `#name` marker in the template giving access to an element or directive instance.

l. Practice Exercises

1. Build a `tabs` wrapper that projects two panels.
2. Create a modal wrapper with `header`, `body` and `footer` slots.

3. Use `viewChild` to scroll a `div` to the top when a button is clicked.